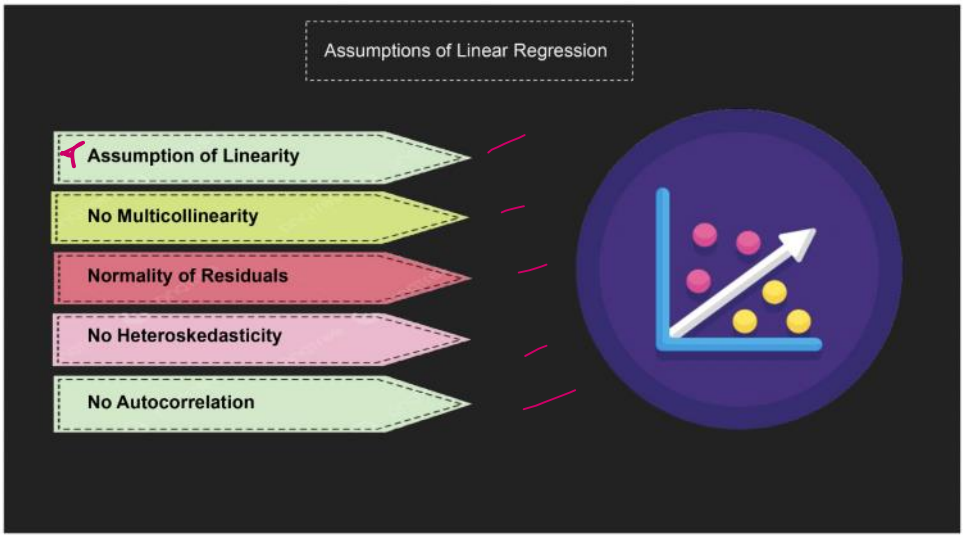
LINEAR Regression — 4

Agenda ⇒

- → Assumptions of Linear Regn
- → VIF → Variance inflation factor
- Heteroskedasticity ⇒
- Autocorrelation



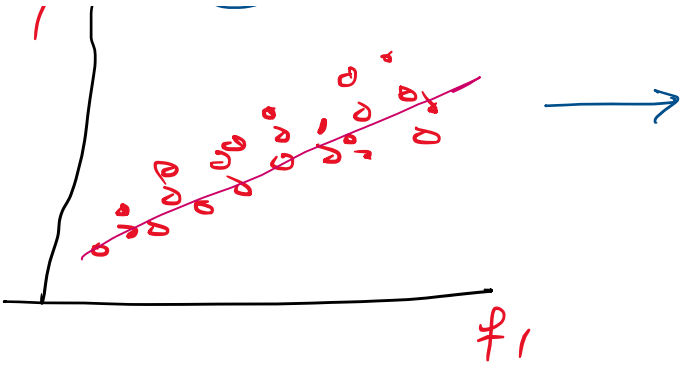
→ LR → poorly ⇒

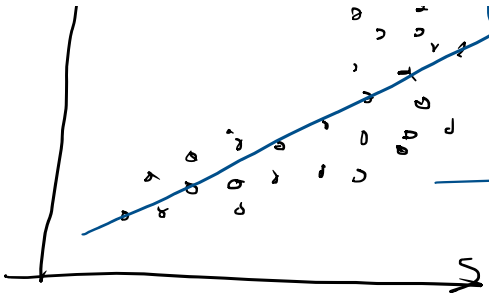
⇒ NON LINEAR

1) Linearity ⇒

I

II → Linear model





→ NON LINEAR

DATA ⇒

→ NO MULTICOLLINEARITY

(*) COLLINEARITY ⇒ $2 \times f_1 + 3$

$2f_1 + 3 = f_2$

(*) f_1, f_2 ⇒ COLINEAR

scalar scalar

if $f_2 = \alpha \cdot f_1 + \beta$

$f_1 \& f_2$ ⇒ ⇒ COLLINEAR

(*) $f_2 = 2 \cdot f_1 + 3$

CARS24

(*) CAR 24

(*) age, year

CARS24 ⇒

[age, years] ⇒ COLLINEAR

2026 − 2024 = 2

① age = $-1 \cdot$ year + 2025 ⇒

② age, year ⇒ collinear

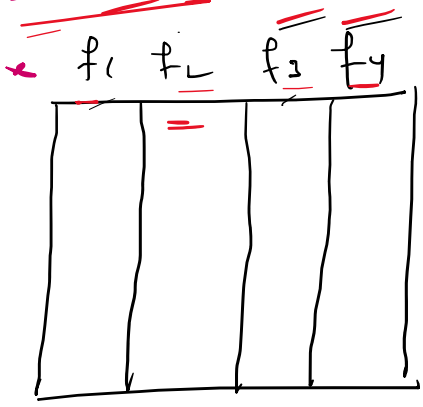
③ Another example → Height in ft
→ Height in cm

④ kg → grams

⇒

* collinearity across multiple features
↳ multicollinearity ⇒

DATA



⇒

4 features

⑤ $f_2 = \alpha_1 f_1 + \alpha_3 f_3 + \alpha_4 f_4 + \alpha_0$ (constant 1, constant 2)

⇒

True (f_2 is multicollinear)

⑥

(2) f_2 → multicollinearity

Transformation

f_2 → using combⁿ of existing features

f_2 → dependent on other features

→ multicollinearity

Cases

(1) $\alpha_3 = 0$ ⇒ f_2 →

$\alpha_1 = -0.8$, $\alpha_4 = 4.1$

$\alpha_5 = 2$

f_1 & f_4

f_3 → not contributing

f3 → not contributing

→ multicollinearity still exists

f2 → No multicollinearity

↳ assumption given by stats people

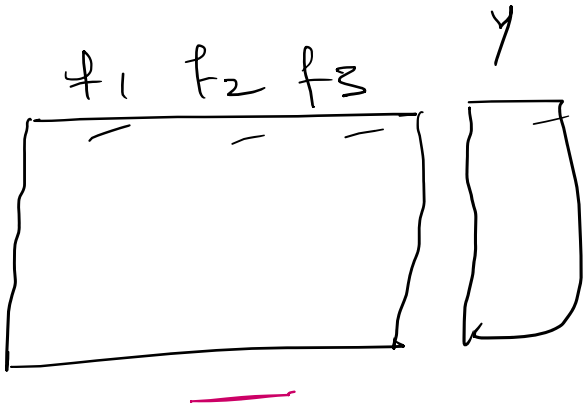
⇒ multicollinearity ?

$$f_1 = \alpha \cdot f_2 + \alpha_0 \quad (*)$$

Linear combⁿ

MC → Problem ?

→



→ ML algo

↓

LR ⇒

↓

* weights → Biases

⊕ Optimization ⇒ w^*

⇒ $[w_1, w_2, w_3]$ w_0 ×

* $w^* = [1, 2, 3]$ $w_0 = 5$

↓

more imp

Test query ⇒

⊕ query DF → $[x_{q1}\ x_{q2}\ x_{q3}]$

⇒ Pred^n

$\hat{y} = 1 \cdot x_{q1} + 2 \cdot (x_{q2}) + 3 \cdot x_{q3} + 5$

$\hat{y} \Rightarrow 1.x_{q_1} + 2.x_{q_2} + 3.x_{q_3} + 5$ → (1)

f_2 f_1

(*) if MC exists ⇒

$f_2 = 1.5\, f_1$ ⇒ (40) ⇒ Decided ⇒ given

$\hat{y} = 1.x_{q_1} + 2(1.5 x_{q_1}) + 3.x_{q_3} + 5$

(*) $\hat{y} = 4x_{q_1} + 3x_{q_3} + 5$ → (2)

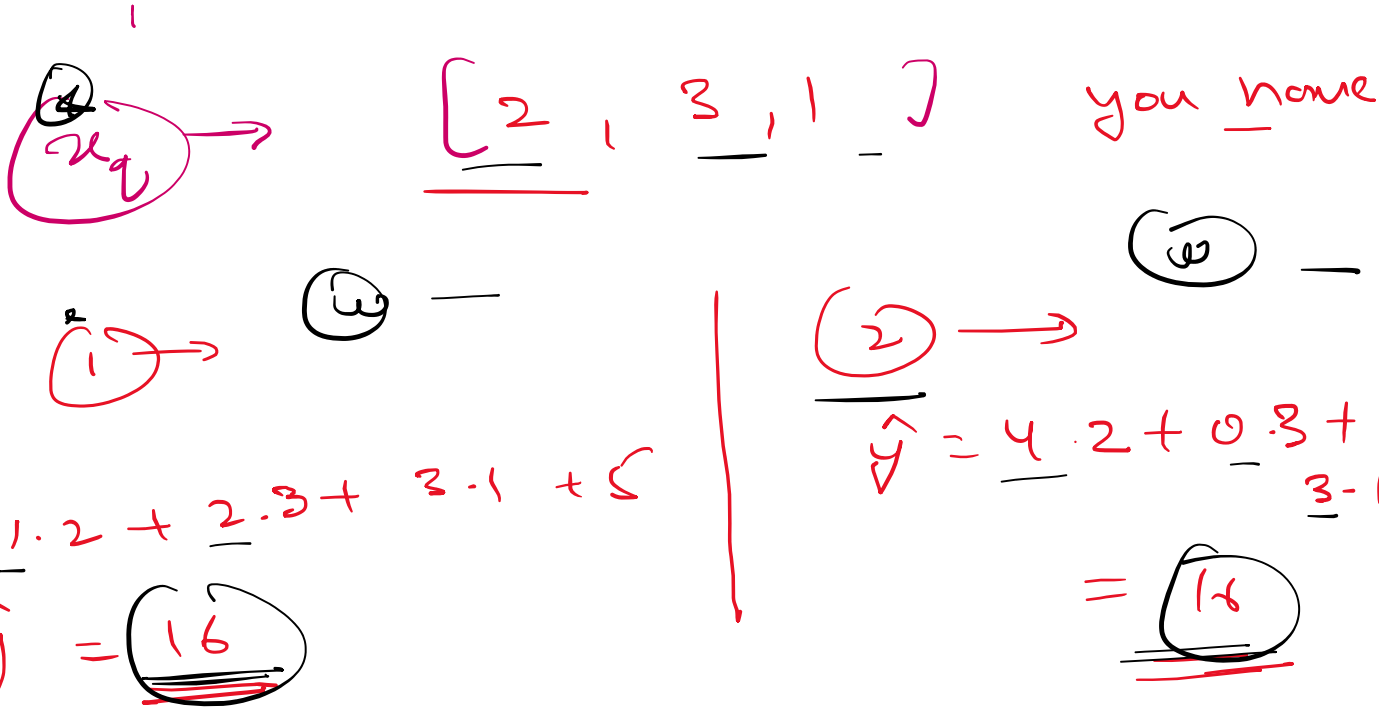
(1) → eqⁿ

$w = [1, 2, 3]$

$\hat{y}$ ⟹ same

(2) → eqⁿ

$w = [4, 0, 3]$ (f_1)



(x) wts → massd → problem

↓

(y) feature imp ✗

wts → unstable → No feature imp

* interpretability — ↓ Challenges

MC → problems

Pred

No multicollinearity

↓ prob & stats

MC → check ⇒

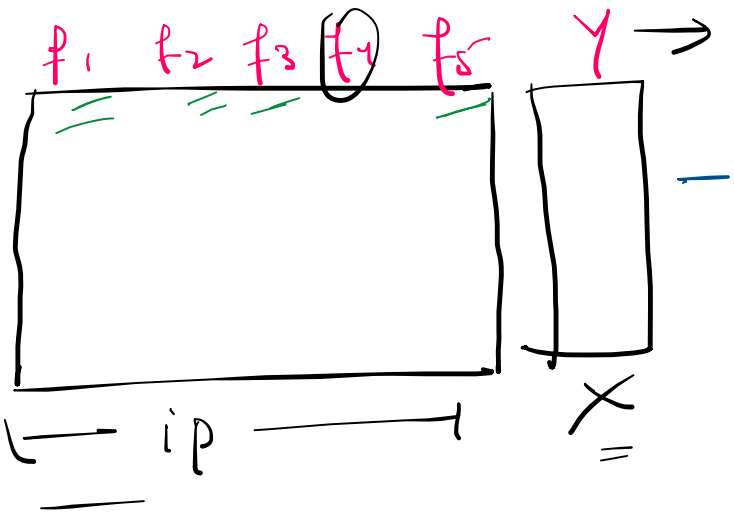
Correlation ⇒ 2 variables

f2

VIF → Variance inflation factor

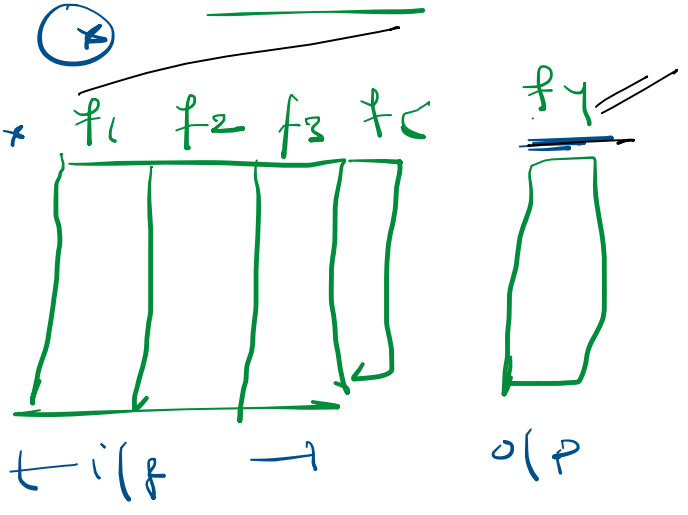
→ Y → ignored

→ Y → ignored



* f4 →

VIF of f4



→ Fit LR model

⇒ f4 = w1f1 + w2f2 + w3f3 + w5f5 + w0

MC exists

(*) R2 - Score ⇒

* 0.18 → worst model

* X

Poor

= incapable of predicting fy

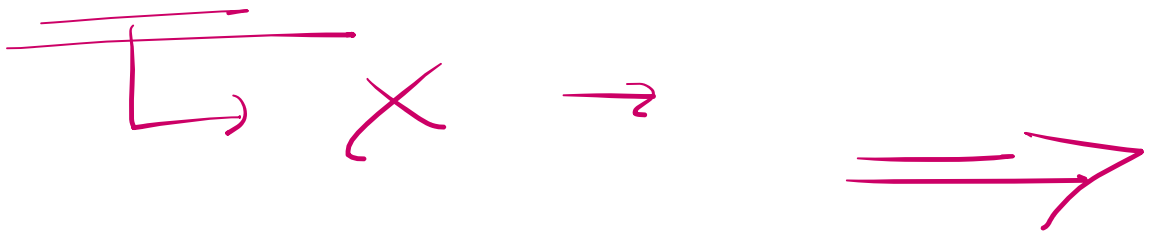
fy → cannot be represented using existing features

MC → does not exists

R2 - score = 0.92 → good model

MC exists

fy → depends



VIF SCORE

$$VIF_{fj} = \frac{1}{1 - R_{2j}} \Rightarrow VIF$$

R_{2j} → R2 score of jth features

if $R_2 = 0$ ✗ $\Rightarrow VIF = 1$

$R_2 = 1 \Rightarrow VIF = \infty$ ↓

(+) large no

NO MC ← → MC exists

VIF ∈ [1, ∞)

MC exists

THUMB RULE + THEORY

- VIF f1
- VIF f2
- VIF f3

VIF > 10 → very High → MC

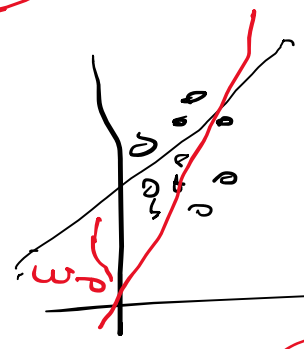
> 5 → MC

< 5 → NO MC

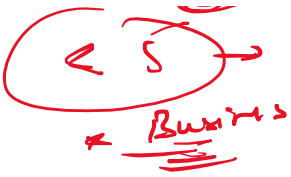
Break → 5 mins → 10:12

Code →

Maths



< 5



Normality of Residuals ⇒

→ ERRORS = $y - \hat{y}$

(+) $y = \underbrace{w^T x + w_0}_{\hat{y}} + \epsilon$

Residual

100%

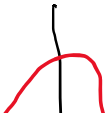
$y = \hat{y} + \epsilon$

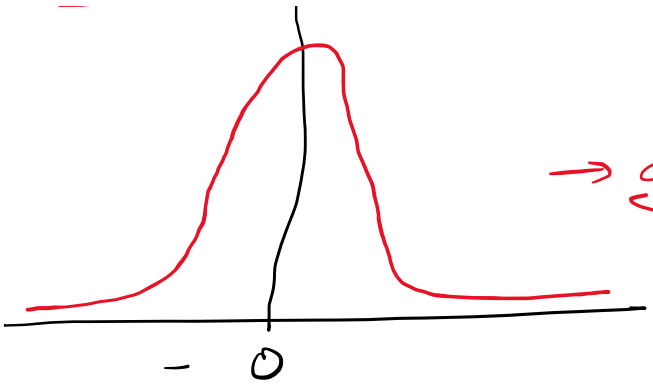
$\boxed{\epsilon = y - \hat{y}}$ ⇒ Residuals

(+) $e \to 0$ X

Residuals → dist^n of errors → for every DP

plotting → Stats





→ gaussian (Norr dist)

↓

centre around 0

=

er → 0

ERRORS → −ve

CARS 2Y

$y = 3.2$ L $\quad \hat{y} = 4.2$ L

$y - \hat{y} \Rightarrow$ −ve ERROR

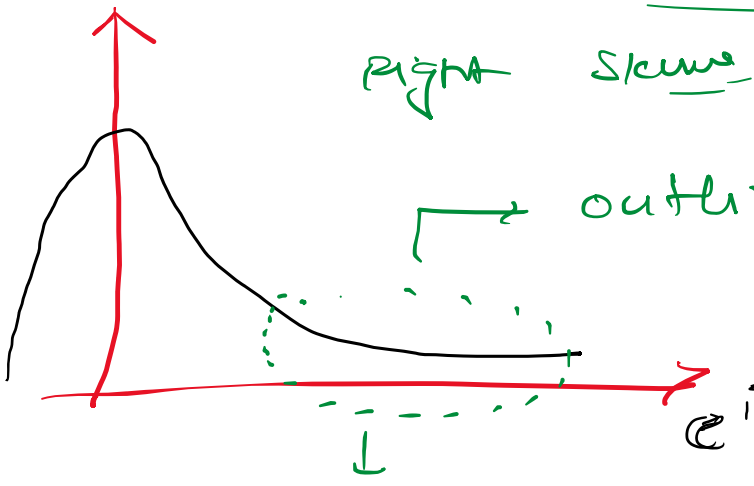
over pred

−ve error

under pred

+ve error

→ under predicting model



Right Skew -

→ outliers → Huge Errors

⇒

few DP → large +ve errors

over predicted model

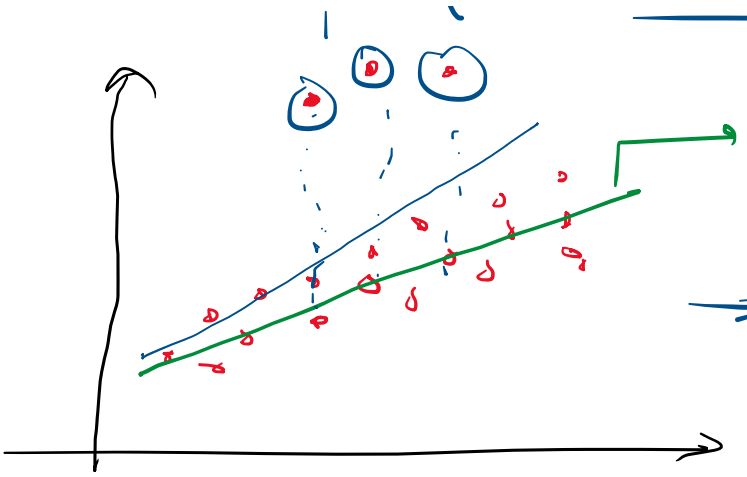


⇒

← Plot of error dist^n

→ model will be inaccurate

→ Huge errors



Best model

→ model is not accurate

impact of outliers

(1) identify outliers

(2) Deal with outliers → Clipping
→ if
- DBSCAN,
- Isolation Forest

⇒ qq plot etc

(3) Normality → Shapiro

errors → Not normal

Residual ERROR analysis

Normal → log / Box cox trans

A clothing store wants to predict sales based on factors like price, promotions, and store location.
Which assumption of linear regression is important for accurate sales predictions?

20 users have participated

A	Linearity between the independent variables and sales.	15%
B	Normal distribution of sales.	5%
C	MultiCollinearity among features	15%
D	All of them	65%

End Quiz Now

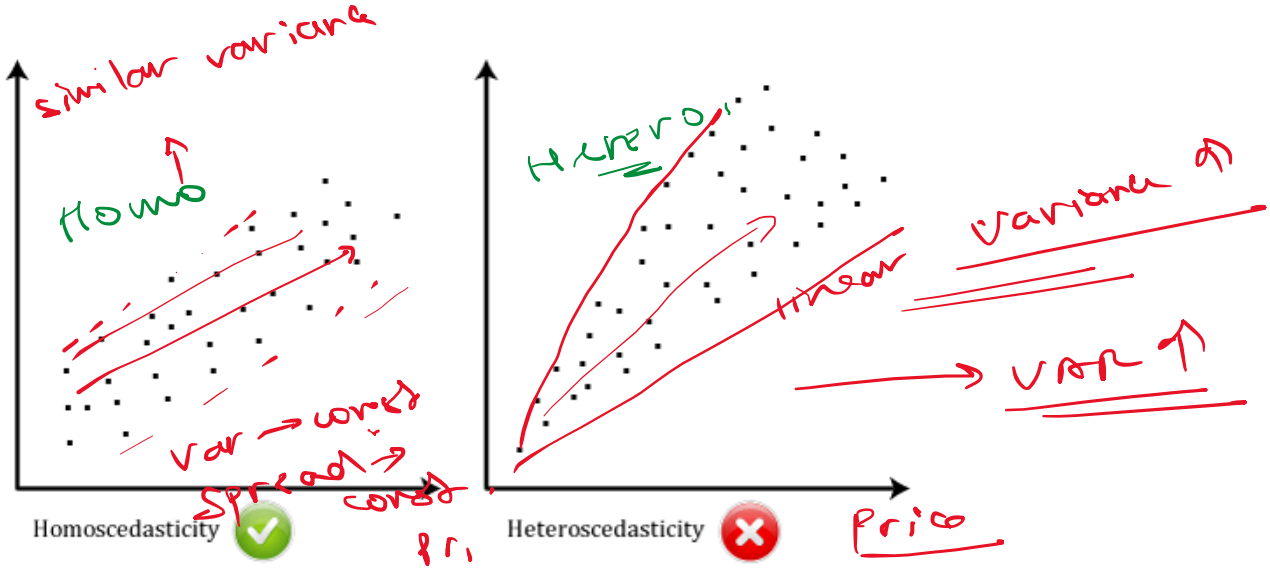
Last assumption → HETERO

NO HETEROSCEDASTICITY

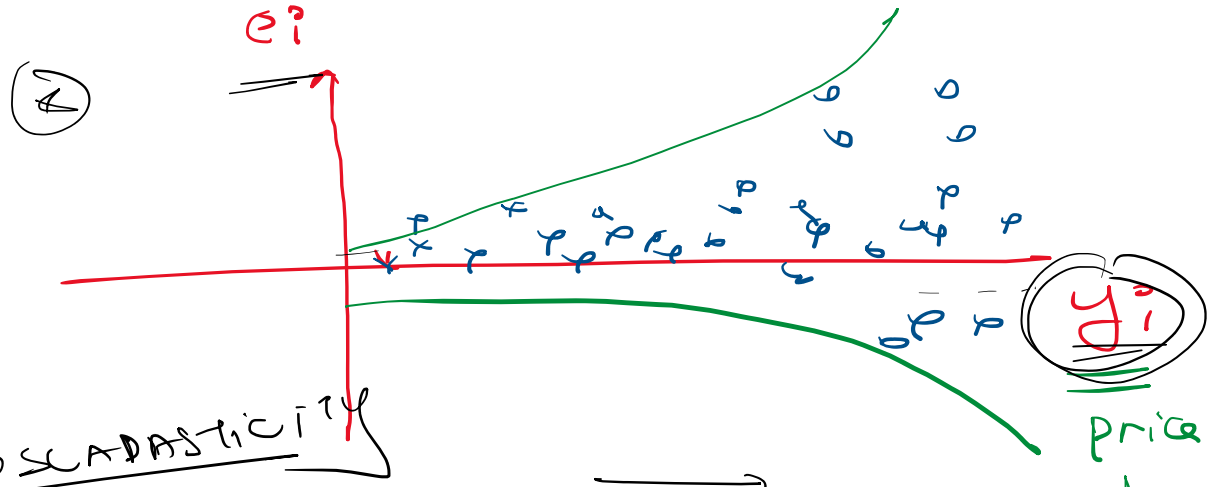
NO HETEROSCADASTICITY

→ Homoscadasticity

$E^{(i)}$ Vs (y_i)



$E^{(i)}$ Vs y_i



HETEROSCADASTICITY

HETEROSCADASTICITY → price of car

Car → BMW, Audi → more variance

70 lac

10 lac

30 lacs

HETI

Cheta → 19

18, 20

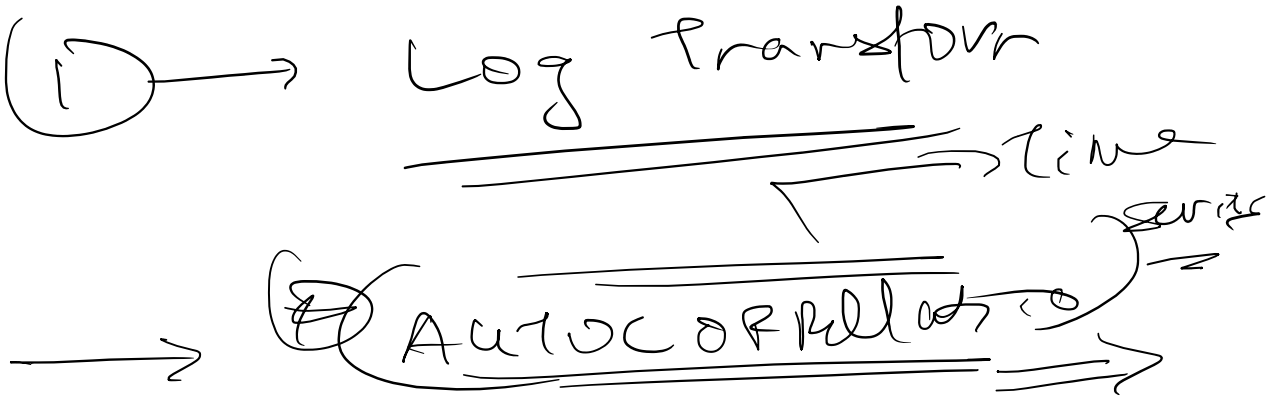
12, 24

No HETERO

→ model → worse

→ impacts goodness of fit

Transformation


1
Log transform
time
series
2
Autocorrelation